

Numerical parameter studies for the dense plasma focus

This content has been downloaded from IOPscience. Please scroll down to see the full text.

1975 Plasma Phys. 17 237

(<http://iopscience.iop.org/0032-1028/17/4/001>)

View [the table of contents for this issue](#), or go to the [journal homepage](#) for more

Download details:

IP Address: 128.119.168.112

This content was downloaded on 01/02/2015 at 14:12

Please note that [terms and conditions apply](#).

NUMERICAL PARAMETER STUDIES FOR THE DENSE PLASMA FOCUS

M. TRUNK

Institut für Plasmaforschung, Stuttgart University, Germany

(Received 25 February 1974)

Abstract—The magnetohydrodynamic equations are solved using an explicit numerical scheme for the focus geometry together with the electrical circuit equation. The influence of varying circuit parameters, focus apparatus dimensions, and filling pressure on the discharge characteristics, especially the maximum current, and the plasma variables in the pinch phase are examined and compared with experimentally determined neutron outputs. An experimentally derived scaling law for the dependence of maximum neutron output on bank energy, filling pressure, and length of the inner electrode is confirmed by the results of the MHD computations.

1. INTRODUCTION

SINCE the conception of the plasma focus device by FILIPPOV *et al.* (1962) and MATHER (1964), a large number of experimental investigations of this phenomenon have been carried out. These experiments show the production of hot plasma of very high density (electron density $n_e = 2\text{--}4 \times 10^{19} \text{ cm}^{-3}$, electron temperature $T_e \sim 2 \text{ keV}$, according to PEACOCK *et al.*, 1972). The compressed plasma column has a length of about 1 cm and 1 mm diameter and emits pulses of hard and soft X-radiation as well as a neutron pulse during its initial lifetime of 50–100 nsec. Even today, the main goal of investigations remains the interpretation of the neutron production mechanism. Part of the desired information is obtained from measurements of neutron energy distribution and neutron flux anisotropy, from which it is hoped to derive the velocity distributions of the ions. It now seems certain that the second, very intense neutron pulse from devices of the Filippov type originates from reactions among thermal ions (MAISONNIER *et al.*, 1972), while with the Mather-type there is only one such pulse and the neutrons probably come from reactions of highly accelerated ions. If the production mechanisms are known, the neutron yield can be increased by appropriate methods.

Thus, while intensive efforts are being made to clarify the phenomena occurring in the compressed plasma focus, there has been relatively little systematic research on the requirements for achieving a focussed plasma as hot and as dense as possible. It is generally emphasized that for good focussing and large neutron yield it is important to have a large current at the beginning of the compression phase (MATHER, 1971). However, in dimensioning the plasma focus devices only the acceleration length is determined such that the current maximum occurs at the end of the acceleration phase. Furthermore, the filling pressure of deuterium is adjusted for maximum neutron yield. These were the important criteria in the design of most devices, with relatively little or no consideration given to variation of electrical and/or geometrical parameters. A variation limited to electrical parameters was published by BERNARD *et al.* (1970) which, however, omits optimizing the filling pressure. Thus, as the authors state, they were unable to derive a comprehensive scaling law. On the other hand, parameter variations are readily carried out theoretically, provided a suitable model for the focus discharge is employed. Such calculations have already been performed by various authors using the simple snow-plough model (PATOU

et al., 1968; RAPP and TRUNK, 1971; VICHREW, 1973). They yield qualitative recommendations for a current maximum as high as possible, depending on the electrical data of the capacitor circuit. However, in this method the dynamics of the plasma layer is not very rigorously described as, among other factors neglected, the flow-off along the layer is not considered. No information on the thermodynamic properties of the plasma can be obtained. On the other hand, the snow-plough model permits, by reducing the equations to dimensionless form, an essential reduction in the number of parameters so that the interrelationships become more understandable (ERZ, 1972). By using an MHD model, however, immediate information on the dynamic as well as the thermodynamic properties of the plasma are obtained.

The basic equations of magneto-plasma-dynamics are essentially nonlinear and the problem is at least two-dimensional. This necessitates numerical methods of solution. The difficulties arising in this were satisfactorily solved for the first time by ROBERTS and POTTER (1970). With their code they were able to describe theoretically the focus discharge up to the pinch phase. The results thus obtained were directly compared with experiments by MORGAN and PEACOCK (1969) and showed good agreement with the plasma data. As in this code, the basic equations of magneto-plasma-dynamics were used in the form described also elsewhere (POTTER, 1971; VEZIN, 1971; TRUNK, 1973). The method of solution used was a Lax-Wendroff scheme (RICHTMYER and MORTON, 1967) with a smoothing procedure according to FREEMAN and LANE (1968). It is the aim of these investigations to optimize the electrical parameters of the energy supply and the geometry of the focus device for maximum current and maximum utilisation of the stored energy and to obtain simultaneously very high density and temperature in the focussed plasma. The conditions achieved in the plasma focus then can be compared with the neutron pulses yielded by the focus devices of RAPP (1973) and PROSS (1973). In the calculations, the relatively coarse grid employed seems to yield plasma variables somewhat too low. However, the advantage of this coarse grid was the possibility of calculating a large number of variations within a reasonable time, so that it was possible to study the changes of the variables for a given time in the calculation. On the other hand, the dependence of the plasma variables on the grid size has been investigated by VEZIN (1971).

2. DESCRIPTION OF DEVICES AND PARAMETERS INVESTIGATED

Since 1971 two focus devices of the Mather-type have existed at the Stuttgart plasma institute. These devices were designed from the very beginning in such a way as to permit variations of the power supply circuit and the discharge geometry. The data for these two installations "MINIFOCUS" and "NESSI" are given in TABLE 1.

The principal behaviour of the focus discharge (formation of a shock wave with constant propagation velocity in the axial phase, rapid collapse forming a hot and very dense focus plasma in the radial compression phase) has already been described in detail by POTTER (1971). This procedure is not essentially different for the two Stuttgart devices (TRUNK, 1973). As the capacitors are somewhat slower than those of the device described by POTTER, the compression velocities are expected to be slower, resulting in lower temperatures for the focus plasma.

The MHD equations are coupled by the magnetic field behind the plasma layer (determined by the discharge current) and by the variable inductance L_1 of the

TABLE 1.—DIMENSIONS AND INITIAL CONDITIONS

	Minifocus	Nessi
Length of inner electrode	$l = 24.5 \text{ cm}$	21–30 cm
Length of insulator	$a = 1.7 \text{ cm}$	3 cm
Radius of inner electrode	$r_i = 2.5 \text{ cm}$	3.3 cm
Radius of outer electrode	$r_a = 4.5 \text{ cm}$	6.25 cm
Capacity of capacitor bank	$C = 61.6 \text{ F}$	225 F
Charging voltage	$U_0 = 20 \text{ kV}$	16–20 kV
Inductance of external circuit	$L_0 = 60 \text{ nH}$	20 nH
Resistance of external circuit	$R = 1 \text{ m}\Omega$	2 m Ω
Filling pressure	$p_0 = 1 \text{ Torr}$	0.5–5 Torr
Initial temperature for calculation	$T_0 = 2 \text{ eV}$	2 eV

accelerator (established by the down-streaming plasma) to the oscillation equation for the electrical circuit,

$$U_0 - \frac{1}{C} \int I dt = (L_0 + L_1) \dot{I} + L_1 I. \quad (1)$$

The solution of this equation for the electric oscillating circuit without plasma focus, and thus the maximum current, is determined by the circuit parameters U_0 , C and L_0 . Changing the parameters in this case would be trivial. However, the oscillation equation is characteristically influenced by the dynamics of the focus discharge through the variable internal inductance $L_1(t)$ of the coaxial accelerator. The solution of the MHD equations describes a plasma layer propagating in the (axial) z -direction, sweeping up the gas upstream. Part of this gas is deflected and flows off to the outside, leaving a vacuum region behind which is filled with magnetic energy. Thus, dL_1/dt becomes proportional to the velocity v of the plasma layer.

These phenomena can be discussed with the aid of Fig. 1 (e.g. curves "2") and a simplified form of the equation of motion,

$$\frac{\partial}{\partial t}(\rho v) = -\text{grad} \left(\frac{B^2}{2} \right) + \dots, \quad (2)$$

where ρ is the density and B the magnetic induction. After ignition, the forming plasma layer at first remains stationary. This has also been shown experimentally (RAPP, 1973). L_1 shows little variation and the current initially follows the undamped oscillation equation. After the current, and thus the magnetic field B_z behind the plasma layer, has reached a certain magnitude, the plasma is accelerated,

$$\left(\frac{\partial v}{\partial t} \right)_a \sim \frac{B_a^2}{\rho_0 \cdot \Delta z} \sim \frac{B_a^2}{m_0}, \quad (3)$$

where ρ_0 is the initial density, Δz the length increment (= layer thickness) of the moving plasma layer, and m_0 is the initial mass of the plasma layer ($\sim \rho_0 \Delta z$). Because of this motion of the plasma layer the current is damped. The critical time for this acceleration process and hence the current reached so far will thus be dependent on the filling pressure $p_0 \sim \rho_0$ and on the radius r_i of the inner electrode, since on the anode $B_a \sim I_a/r_i$. After this, the inductance $L_1(t)$ increases linearly and immediately a constant propagation velocity is established, for which

$$dL_1/dt = \frac{d}{dt} \int_{r_i}^{r_a} \int_{z=0}^z \frac{1}{r} dr d\zeta = f \cdot \frac{dz}{dt} = f \cdot v_a. \quad (4)$$

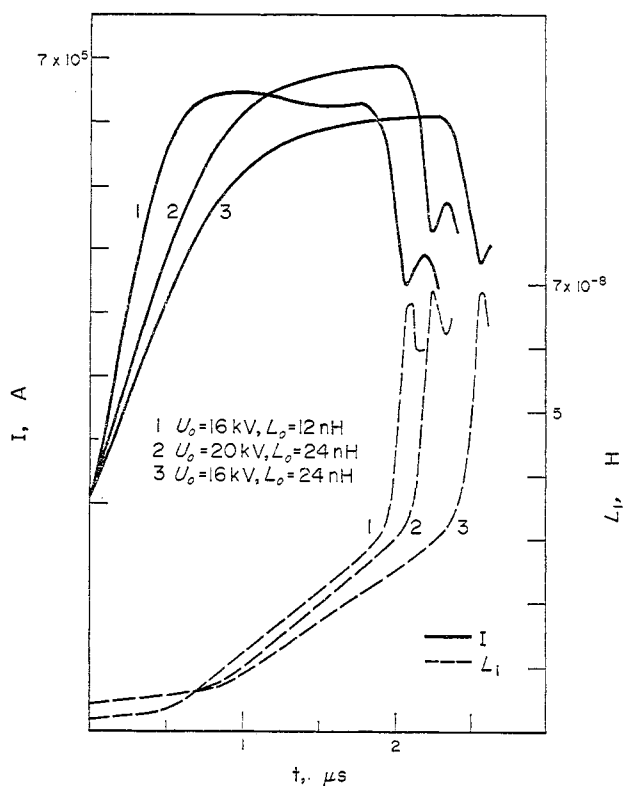


FIG. 1.—Discharge characteristics for “NESSI” at various capacities charging voltages U_0 and various external inductances L_0 .

From the equation of motion,

$$\frac{\Delta \rho}{\Delta t} \cdot v_a \sim \frac{B_a^2}{\Delta z} \rightarrow v_a^2 \sim \frac{B_a^2}{\Delta \rho}, \quad (5)$$

it is seen that the damping rate $dL_1/dt \sim v_a$ is again dependent on the filling pressure, as the change of mass $\Delta \rho$ produced by sweeping up the stationary gas upstream of the plasma layer is proportional to the filling pressure. Furthermore, dL_1/dt will depend on the ratio of the electrode radii r_a/r_i . These considerations apply analogously also to the current dip in the radial collapse phase.

However, the actual amount of this dependence of the current on the discharge mechanism is also influenced by the ratio of external to internal inductance L_0/L_1 . For large L_0 , the discharge will be more resistant to current dips caused by L_1 than for small L_0 . On the other hand, maximum current will be lower for larger L_0 , as can be seen from the oscillation equation. Furthermore, the discharge time τ of the capacitor bank for given bank energy E_0 ,

$$\tau \sim \sqrt{L_0 C} \sim \frac{\sqrt{L_0}}{U_0}, \quad (6)$$

will play a certain rôle. For a fast bank with rapid current rise $\dot{I}(0) = U_0/L_0$, the current will already have increased to a larger value before the plasma layer starts to move and thus damp the current, the motion of the layer being retarded by its inertia.

In the following, the influences of the parameters, considered important from the above approximate analysis, will be investigated quantitatively by a numerical solution of the MHD equations in conjunction with the oscillation equation for the current circuit. It appears reasonable to separate the parameters into those of the external circuit and those of the discharge.

3. PARAMETRIC VARIATION OF THE ENERGY SUPPLY

a. Charging voltage U_0 .

The simplest way of increasing the neutron yield from the focus discharge is to increase the maximum current. According to MATHER *et al.* (1971), the neutron output varies as $I_{\max}^4$. As $I_{\max} = \sqrt{2E_0/L}$ ($L = L_0 + L_1$), it seems at first that an increase of current for given capacitor energy E_0 can only be achieved by lowering the inductance L and that it would make no difference whether a bank with low charging voltage U_0 and large capacity C is available or vice versa. However, the choice will prove important in the concept of a focus bank, as the electrical oscillation circuit is strongly influenced by the dynamics of the focus discharge. Thus, the short-circuited 45 kJ bank of the Stuttgart plasma focus "NESSI" with its external inductance L_0 of 24 nH would yield a maximum current of 1.76 MA, while the calculations yield only 0.8 MA for a charging voltage of 20 kV. With 100 kV charging voltage and the same energy, a theoretical maximum current of 1.4 MA would result. The length of the electrodes will of course have to be optimized for the different discharge times $\tau \sim \sqrt{L_0 C}$ ($l = 18$ cm for 20 kV and $l = 7.2$ cm for 100 kV). The shorter discharge tube for higher voltage has a lower internal inductance L_1 and thus implies less damping of the current. The essential effect is, however, that in the case of the fast bank the current increases faster than the plasma layer, due to its inertia, can move, so that motion of the layer occurs when the current is already quite large. Acceleration of the plasma layer commences after $0.8 \mu\text{s}$ at a current of 0.6 MA for 20 kV charging voltage and after $0.27 \mu\text{s}$ at 1 MA for 100 kV.

b. External inductance L_0 .

Another way to make a capacitor bank faster is by reducing its inductance L_0 . The characteristics of the discharge for $L_0 = 1.2 \times 10^{-8}$ and $L_0 = 2.4 \times 10^{-8}$ H for a charging voltage of 16 kV are given in Fig. 1. As expected, the rate of rise of the current $\dot{I}(0) = U_0/L_0$ is twice as large for half the inductance L_0 . In the short-circuited case, $I_{\max}$ is expected to be a factor of $\sqrt{2}$ higher for the lower inductance. However, the more rapid initiation of the plasma acceleration, evident by the earlier rise of L_1 , immediately damps the current so that a plateau is reached which is significantly lower than this predicted value. The quite unfavourable ratio of external to accelerator inductance influences the oscillation equation so strongly that the current curve even shows a marked dip. Although for lower inductance L_0 the current at the beginning of the compression is higher, the current dip for the compression phase is much larger because the inductively stored energy is only about half as large as for $L_0 = 2.4 \times 10^{-8}$ H. In this way, the net effect of the force of the compressing magnetic field is equal in both cases and the plasma variables are almost identical in the compressed stage. The situation is worse for still smaller inductances. The best conditions are achieved in a third calculation for $L_0 = 1.8 \times 10^{-8}$ H. However, the variation of the optimum is quite flat within the range of L_0 considered in these

calculations, as has already been shown by snow-plough calculations (RAPP and TRUNK, 1971). All in all, it can be stated that the optimum conditions for the focus experiment "NESSI" occur for an external inductance L_0 of about 20 nH.

c. Capacitor energy E_0 .

Also shown in Fig. 1 are the discharge characteristics for two different capacitor energies obtained by varying the charging voltage from 16–20 kV. The resulting differences are readily seen: Higher voltage causes a faster rate of current rise and higher maximum current, as $I(0) \sim U_0$ and $I_{\max} \sim U_0$. A larger rate of current rise means a greater axial propagation velocity for the plasma layer. The current ratio $I_{\max}(16 \text{ kV})/I_{\max}(20 \text{ kV}) = 0.88$ is clearly reflected in the plasma variables, as can be seen from Table 2.

TABLE 2.—PLASMA VARIABLES AT MAXIMUM FOCUS COMPRESSION FOR VARIOUS CHARGING VOLTAGES AT CONSTANT PRESSURES AND CONSTANT ELECTRODE LENGTHS

	16 kV	20 kV	X(16 kV)/X(20 kV)
Time	2.558	2.239 μs	
Density at centre of anode	3.44	$3.85 \times 10^{19} \text{ cm}^{-3}$	0.89
Magnetic field strength at a radial distance of 0.5 cm from axis	1.44	$1.88 \times 10^5 \text{ Gauss}$	0.80
Electrical field strength at centre of anode	1.79	$2.26 \times 10^4 \text{ V/cm}$	0.79
Current density at centre of anode	4.82	$6.07 \times 10^5 \text{ A/cm}^2$	0.79
Electron temperature on axis	3.26	$4.13 \times 10^2 \text{ eV}$	0.79
Ion temperature on axis	3.32	$4.28 \times 10^2 \text{ eV}$	0.79

4. VARIATION OF PARAMETERS OF THE DISCHARGE VESSEL

a. Internal inductance L_1 .

From the foregoing it is seen that the internal inductance is especially detrimental to high maximum current. It may thus be desirable to influence L_1 directly. As $L_1 \sim \int_{z=0}^z \int_{r_i}^{r_a} 1/r \, dr \, dz$, this requires a change of the focus geometry. Changing the length is trivial, as the plasma layer must have reached the end of the inner electrode when the current has reached its maximum. A reduction of the inductance per unit length is achieved by reducing the radius ratio r_a/r_i (for a plane plasma layer $dL_1/dz \sim \ln r_a/r_i$). In order to ascertain the dependence of the focus discharge on the internal inductance, examples for two different inner radii r_i with constant outer radius $r_a = 6.2 \text{ cm}$ were calculated. The charging voltage was 16 kV in both cases. The results are presented in Table 3.

TABLE 3.—DEPENDENCE OF A FOCUS DISCHARGE ON THE RADIUS OF THE INNER ELECTRODE

		r_i'	r_i''	X''/X'
$L_0 = 2.4 \times 10^{-8} \text{ H, 1 Torr}$	In (r_a/r_i)	3.34	4.3 cm	1.3
	$I_{\max}$	0.62	0.37	1:1.7
	dL_1/dt	0.60	0.80 mA	1.33
	$\rho_{\max}$	1.60	$0.90 \times 10^{-3} \text{ H/s}$	1:1.78
	$I_{\max}$	2.69	$4.39 \times 10^{19} \text{ cm}^{-3}$	1.64
	dL_1/dt	0.70	0.88 mA	1.25
$L_0 = 1.2 \times 10^{-8} \text{ H, 1 Torr}$	$\rho_{\max}$	1.26	$0.73 \times 10^{-3} \text{ H/s}$	1:1.73
	$I_{\max}$	5.14	$7.29 \times 10^{19} \text{ cm}^{-3}$	1.42
	dL_1/dt	0.65	0.86 mA	1.33
	$\rho_{\max}$	1.94	$1.19 \times 10^{-3} \text{ H/s}$	1:1.63
	$I_{\max}$	2.57	$4.00 \times 10^{19} \text{ cm}^{-3}$	1.56
	dL_1/dt			

The maximum current does not now change in proportion to the ratio of the internal inductance per unit length $dL_1/dz \sim \ln(r_a/r_i)$ but is only proportional to the ratio of the inner radii r_i''/r_i' . An effective acceleration of the plasma layer begins only after a sufficiently large magnetic field $B_a \sim I/r_i$ has been reached. This magnetic field must be the same for constant pressure and constant rate of current rise, independent of the inner radius r_i . Of importance to the time when the damping of the current begins is not so much the internal inductance per unit length dL_1/dz but rather the time when L_1 starts to increase as soon as the plasma layer begins to propagate. Since the same magnetic field strengths are reached, the propagation velocity of the plasma layer $v_a = dz/dt$ is also the same despite the different radii r_i . The rate of change of the internal inductance $dL_1/dt = (dL_1/dz) \times (dz/dt)$ thus turns out to be proportional to the inductance per unit length $dL_1/dz \sim \ln(r_a/r_i)$. The plasma variables themselves are throughout more favourable for larger inner radius. For example, the density in the focus plasma ρ_{max} increases more than in proportion to the current.

For larger inner radius, the axial acceleration starts later so that in order to make the arrival of the plasma layer at the end of the inner electrode and the occurrence of the current maximum coincide, a shorter inner electrode is required. This means that there is a change from the Mather to the Filippov type of focus. In such a change it must of course be considered that the increase of the inner radius r_i is limited by the value of the magnetic field $B_a \sim I/r_i$ being too low to accelerate the plasma layer. Thus it will turn out that for each focus bank there is an optimum radius for the inner electrode. This optimum radius will be the larger the greater the bank energy. The same conclusion has been reached by IMSHENNIK *et al.* (1973) using a similarity theory.

b. Filling pressure p_0 .

The influence of various filling pressures p_0 on the discharge characteristics of the Stuttgart device "MINIFOCUS" is given in Fig. 2. The critical acceleration $(\partial v/\partial t)_a \sim Ba^2/p_0$ [see (4)] for the beginning of axial plasma propagation obviously seems to be about equal for all three cases, so that during the time of the initial increase in L_1 there is an approximate relation $B_a \sim \sqrt{p_0}$ or, hence, $I_a \sim \sqrt{p_0}$. The axial propagation velocity $v_a \sim (dL_1/dt)_a$ reached thereafter also appears to be proportional to $1/\sqrt{p_0}$, as is evident from (5), so that higher maximum current occurs resulting from the smaller damping at higher densities. The radial compression velocity in the ensuing collapse phase is even more strongly dependent on the filling pressure. The relation $v_a \sim (dL_1/dt)_c \sim 1/p_0$ holds, as shown in Table 4.

For pressures of 0.5 and 1 Torr, the current in the collapse phase shows a first dip with a 20% reduction which has also been experimentally verified to order of magnitude. For a pressure of 2 torr, the current dip is much less pronounced. The device "MINIFOCUS" does not produce a measurable neutron yield for pressures above 2 Torr. This is easily explained by the results of the calculations: the maximum current increases with increasing pressure; however it will asymptotically approach the short-circuit current, which, in the case of "MINIFOCUS", is 5.78×10^5 A. For 2 Torr, the maximum current is already 5.0×10^5 A, so that a further increase in pressure will not produce a significant increase in current. Therefore, the critical condition for acceleration, $B_a \sim \sqrt{p_0}$, is violated for pressures > 2 Torr in

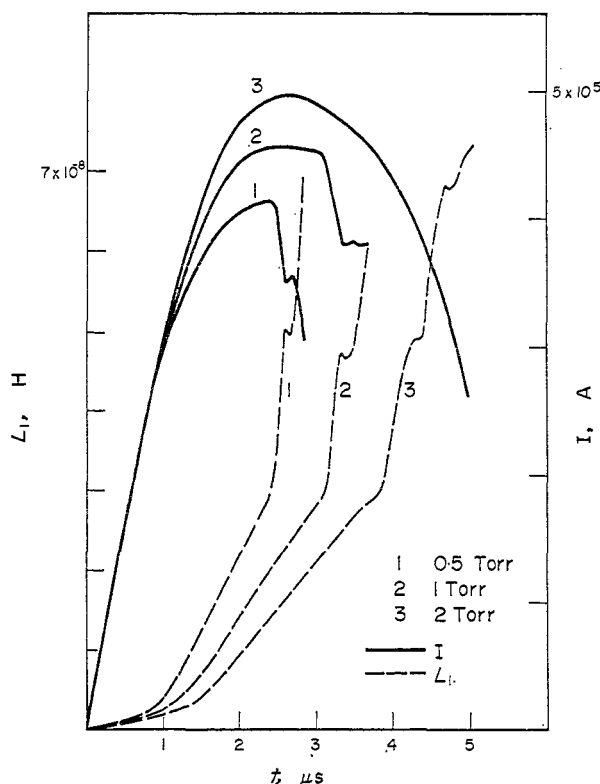


FIG. 2.—Discharge characteristics for "MINIFOCUS" at three different filling pressures.

"MINIFOCUS". As will be shown later, all further plasma variables depend on the collapse velocity achieved by the acceleration.

From the focus device "NESSI", systematic measurements of neutron yield and anisotropy for different filling pressures and lengths of inner electrode are available. In parallel with these experiments, calculations were performed for the pressure range 0.5 to 5 Torr and for lengths of inner electrode 18, 22 and 26 cm, as well as capacitor energies 30 and 45 kJ. The results of these calculations for the various plasma parameters are presented in Figs. 3 and 4; Figs. 5a and 5b show measured neutron yields and anisotropy for the sake of comparison. It turned out that for the range of parameters covered the values of the plasma variables did not change as long as the product of filling pressure times inner electrode length, $p_0 \times l$, remained constant. The explanation is simple: higher pressures mean smaller propagation velocities, so that for the same period of the circuit $\tau \sim \sqrt{L_0 C}$ the inner electrode must be shortened in order that the plasma layer reach the end of the inner electrode

TABLE 4.—TEMPORAL CHANGE OF ACCELERATOR INDUCTANCE DURING THE AXIAL PROPAGATION AND RADIAL COLLAPSE PHASES

Filling pressure p_0		0.5	1	2 Torr	
Rate of change of inductance $dL_i/dt \sim v$	axial	2.0	1.4	1.0×10^{-8} H/s	$1/\sqrt{p_0}$
	radial	20.0	11.1	5.0×10^{-8} H/s	$1/p_0$

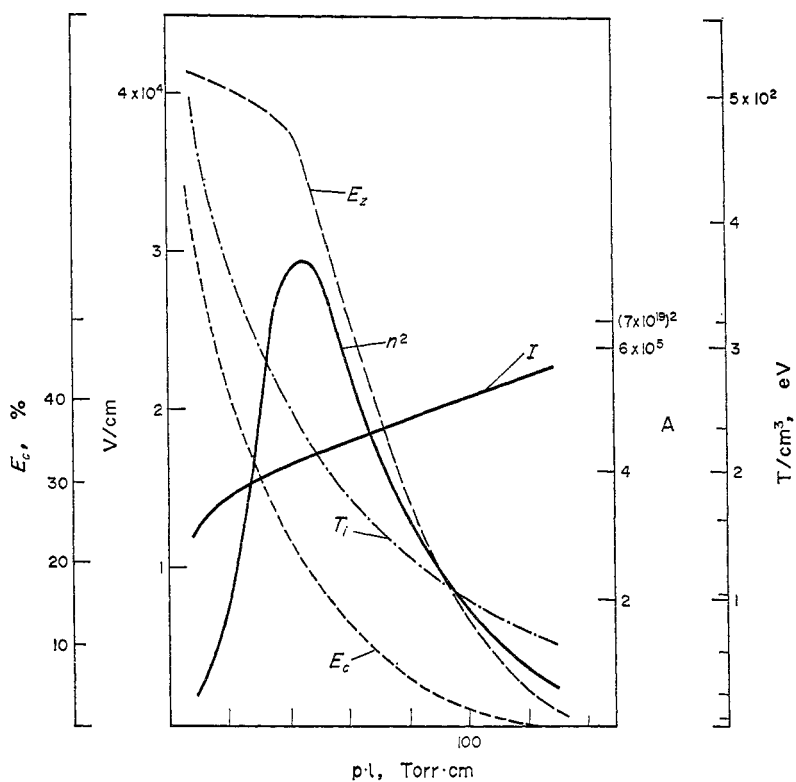


FIG. 3.—Remainder current I , remainder energy E_c and plasma variables for the time of collapse as functions of $p_0 \times l$.

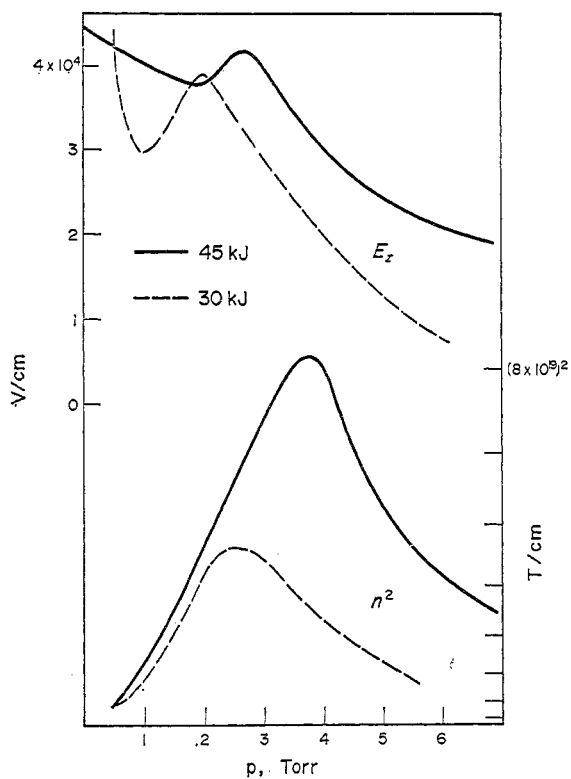


FIG. 4.—Square of particle density and electric field strength on the anode at time of collapse for two different condenser energies.

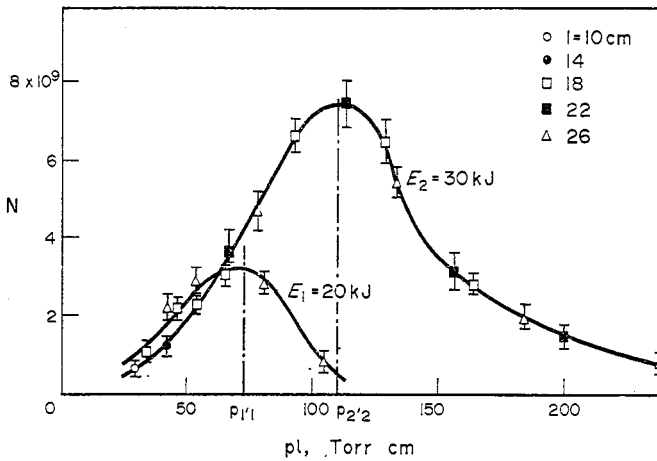


FIG. 5a.—Neutron yield as a function of $p_0 \times l$ at two different condenser energies.

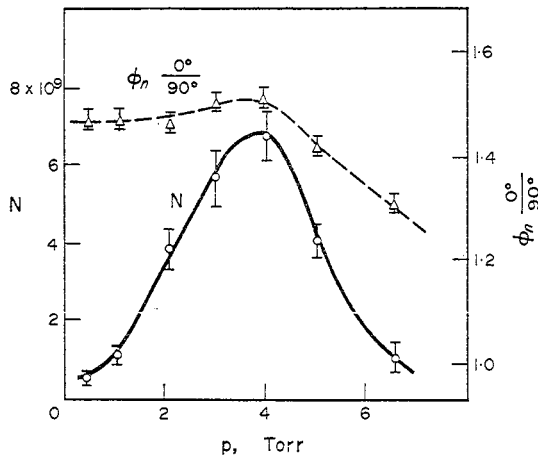


FIG. 5b.—Neutron yield and anisotropy as a function of filling pressure.

just at the time of current maximum. For the neutron yield, the experiment resulted in the same dependence on the product $p_0 \times l$, as shown in Fig. 5a. For these reasons, the plasma variables are plotted against the values of $p_0 \times l$ at the focussing time in Figs. 3 and 4. The remaining current increases with pressure, as the change of inductance in the coaxial accelerator is smaller for higher pressures. Due to the longer propagation time, the capacitor energy E_c is better utilized for larger values of $p_0 \times l$, so that when considering only the effectiveness of the energy supply, high pressures would be quite desirable. However, there is an optimum pressure range for the plasma variables which should not be exceeded. The square of the particle density, n^2 , an essential parameter for determining neutron yield, shows a decided maximum against $p_0 \times l$ similar to the neutron yield itself found in the experiment. While at low pressures the radial collapse velocity, and thus the compression ratio, is quite high, the initial density is too low to attain high final densities. On the other hand, for high pressures the initial density is quite high but the compression ratio too small for high densities in the focussed plasma. Thus, there is

an optimum working range for each focus device depending on the product $p_0 \times l$. In the case of low pressures, the axial electric field is quite strong and seems to permit an acceleration mechanism described by BERNSTEIN (1970) in which a group of ions gain very high energy in a sort of cyclotron motion. Accordingly, strong anisotropy is measured from experiment in this range (see Fig. 5b). Beyond the density maximum towards higher pressures, the electric field strength drops rapidly, while the ion temperature does not decrease as much. This seems to justify the assumption of a Maxwellian ion energy distribution for this range. Along with this, the measured anisotropy of the neutron flux decreases for increasing $p_0 \times l$. The optimum pressure range depends on the capacitor bank energy, as shown in Fig. 4. Hence, the experimentally determined similarity law (Fig. 5a; see also RAPP (1973), RAPP and TRUNK (1973))

$$\frac{E_0}{p_0 \cdot l} = \text{const} \quad (7)$$

has been corroborated by the MHD calculations. Particle densities, electric and magnetic fields as well as temperatures are higher for greater energies E_0 in the optimum pressure range. A dependence for the neutron yield, however, can only be established with the aid of an appropriate neutron production model. This, of course, must wait until particle calculations for the focussed plasma are available.

c. Extension of the Similarity Law

The similarity law (7) may be extended to include an optimum inner electrode radius r_i which depends on the capacitor energy E_0 with the aid of (3) and the comments made in Chapter 3a of this paper. The onset of current damping and thus the maximum current obtainable depends on the critical acceleration of the plasma, thus,

$$\left(\frac{\partial v}{\partial t}\right)_a \sim \frac{B_a^2}{\rho_0} \sim \frac{I_a^2}{p_0 r_i^2} \sim \frac{E_0}{p_0 r_i^2}. \quad (8)$$

For the case in which the critical acceleration is practically constant, as confirmed by the calculations, it follows directly that

$$\frac{E_0}{p_0 \cdot r_i^2} = \text{const.} \quad (9)$$

Thus, if the optimum radius r_i for a given bank energy has been found (e.g. by MHD parameter studies or from technical considerations), the filling pressure p_0 is given and $p_0 \times l$ can no longer be varied according to (7). If the bank energy is increased by increasing the voltage, the oscillation period $\tau_0 \sim \sqrt{L_0 C}$ is independent of E_0 . Aiming at a constant ratio of oscillation period τ_0 to running-down time τ_a of the plasma layer in order to optimize the use of the capacitor energy, it follows from (5) that

$$v_a^2 = \frac{l^2}{\tau_a^2} \sim \frac{B_a^2}{\Delta \rho} \sim \frac{E_0}{p_0 \cdot r_i^2} \quad (10)$$

and, for $\tau_0 = \text{const}$ and $\tau_0/\tau_a = \text{const}$ for $1/\tau_a^2 = \text{const}$, it follows that

$$\frac{E_0}{p_0 \cdot r_i^2 \cdot l^2} = \text{const.} \quad (11)$$

From (9), it now follows that the length of the accelerator l is fixed, independent of bank energy E_0 , as long as $\tau_0 \sim \sqrt{L_0 C}$ is fixed.

These plausibility considerations indicate the important trends for changing the size of focus devices. For each individual case, however, the parameters must be specified as the result of detailed numerical calculations.

5. SUMMARY

In order to choose the appropriate dimensions of a plasma focus device it is essential to perform parameter studies. It is not sufficient merely to increase the capacitor bank energy in some manner in order to achieve a higher neutron yield. In this paper, as well as in that by IMSHENNIK *et al.* (1973), the importance of increasing the bank energy by increasing the voltage U_0 has been shown. That there is an optimum value for the external inductance has also been shown by ERZ (1972) and VICHREW (1973). Similarity laws for optimizing the geometry of the accelerator and the filling pressure have been established here with the aid of MHD theory. In part, these similarity laws await experimental verification from measurements of neutron yield over a range of the respective parameters.

Acknowledgements—The author wishes to thank Prof. Dr. R. WIENECKE for his interest and furthering of this research, and Dr. H. RAPP for the use of Figs. 5a and b.

REFERENCES

- BERNARD A., COUDEVILLE A. and WATTEAU J. P. (1970) *Phys. Lett.* **33A**, ⁸ 477.
- BERNSTEIN M. J. (1970) *Physics Fluids* **13**, 11.
- ERZ U. (1972) IPF-72-4.
- FILIPPOV N. V., FILIPPOVA T. I. and VINOGRADOV V. P. (1962) *Nucl. Fusion Suppl.* Pt. 2.
- FREEMAN J. R. and LANE JR. F. O. (1968) *Proc. APS Topical Conf. Numerical Simul. of Plasma*, Sept. 18–20, 1968, C7-1.
- IMSHENNIK V. S., FILIPPOV N. V. and FILIPPOVA T. I. (1973) *Nucl. Fusion* **13**, 929–934.
- MAISONNIER CH., PECORELLA F., RAGER J. P. and SAMUELLI M. (1972) 5. *Europ. Conf. Contr. Fusion and Plasma Physics*, Vol. 2, Grenoble 21–25 Aug., 1972.
- MATHER J. W. (1964) *Physics Fluids Suppl.* **7**, 5–28.
- MATHER J. W. (1971) *Methods of Experimental Physics* Vol. 9, Pt. B (Edited by R. H. Lovberg and H. R. Griem). Academic Press, New York.
- MORGAN P., PEACOCK N. J. and POTTER D. E. (1969) 3. *Europ. Conf. Contr. Fusion and Plasma Physics* Utrecht 23–27 June 1969.
- PATOU C., SIMONNET A. and WATTEAU J. P. (1968) *J. Phys.* **29**, 973.
- PEACOCK N. J., FORREST M. J., HOBBY M. G. and MORGAN P. D. (1972) 5. *Europ. Conf. Contr. Fusion and Plasma Physics* Vol. 1, Grenoble 21–25 Aug. 1972.
- POTTER D. E. (1971) *Physics Fluids* **14**, 9.
- PROSS G. (1973) private communication.
- RAPP H. and TRUNK M. (1971) IPF-71-1.
- RAPP H. (1973) *Phys. Lett.* **43A**, 5.
- RAPP H. (1973) Thesis, University of Stuttgart.
- RAPP H. and TRUNK M. (1973) 6. *Europ. Conf. Contr. Fusion and Plasma Physics* Vol. 1, Moscow July 30–Aug. 4, 1973.
- RICHTMYER R. D. and MORTON K. W. (1967) *Difference Methods for Initial-Value Problems* 2. edition. Interscience, New York.
- ROBERTS K. V. and POTTER D. E. (1970) *Methods of Computational Physics* Vol. 9. Academic Press, New York.
- TRUNK M. (1973) IPF-73-2.
- VEZIN R. (1971) Centre d'Etudes de Limeil, Rapport CEA-R-4219.
- VICHREW W. W. (1973) *PMTF* (russ.) **2**, 160.